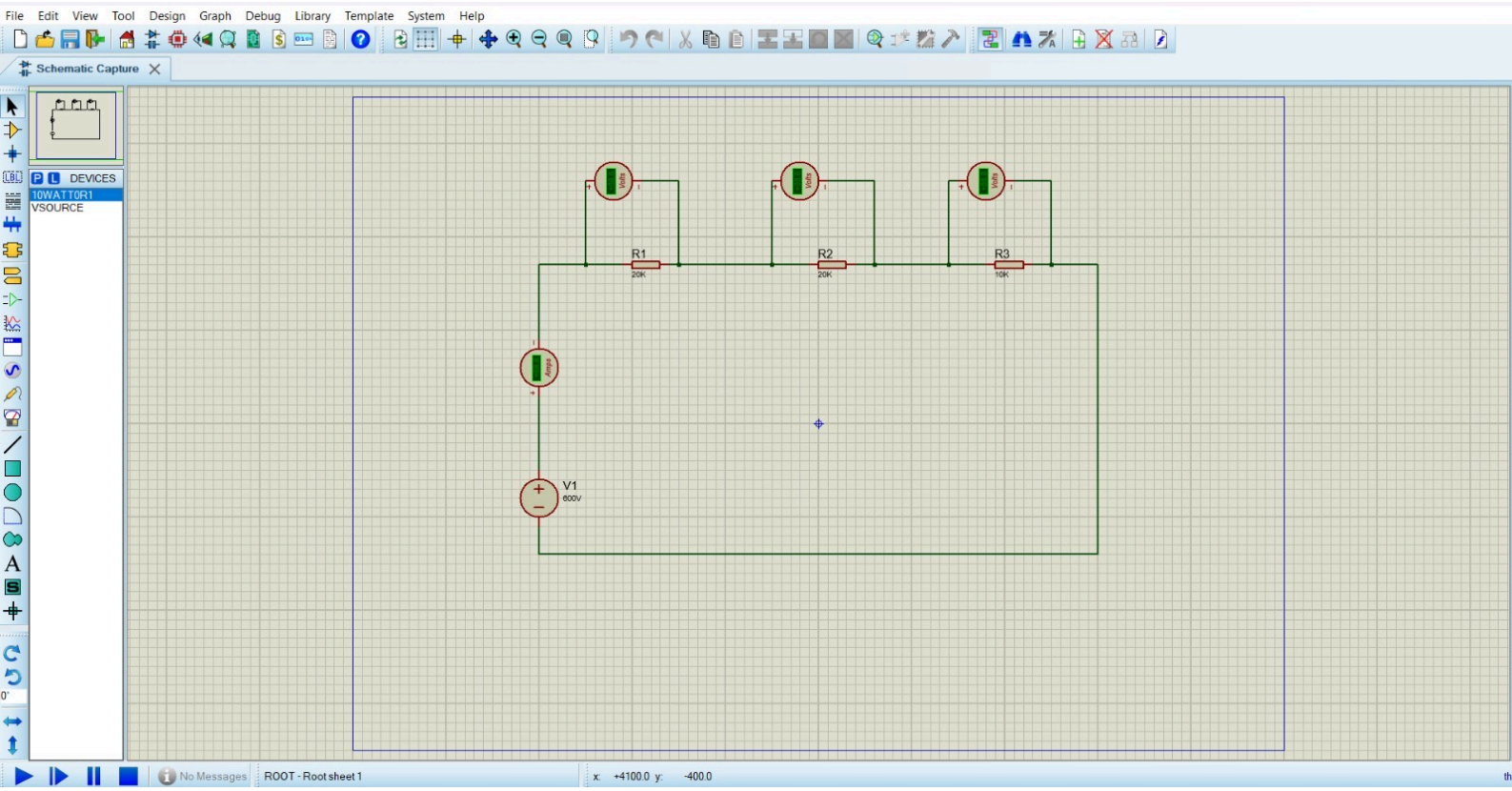

File Edit View Tool Design Graph Debug Library Template System Help
Schematic Capture
DEVICES
10WATT0R1
VSOURCE
R1
20K
R2
20K
R3
10K
V1
600V
No Messages
ROOT - Root sheet 1
x: +4100.0 y: -400.0

KVL EXPERIMENT

MANUAL CHECKING

→ $V = 800V$

→ $R_1 = 20kV = 20000\Omega$

→ $R_2 = 20kV = 20000\Omega$

→ $R_3 = 10kV = 10000\Omega$

→ ~~R_4~~ =

Resistors are in series.

$$R_T = R_1 + R_2 + R_3$$

$$R_T = 20,000 + 20,000 + 10,000$$

$$R_T = 50,000\Omega$$

$$\boxed{R_T = 50k\Omega}$$

Using Ohm's law:

$$I = \frac{V}{R_T}$$

$$I = \frac{800}{50,000}$$

$$\boxed{I = 0.016A = 16mA}$$

Voltage across each resistors,

$V_1 \longrightarrow R_1$

$\Rightarrow V_1 = IR_1$

$= 0.016 \times 20000$

$\boxed{V_1 = 320V}$

$V_2 \longrightarrow R_2$

$\Rightarrow V_2 = IR_2$

$= 0.016 \times 20,000$

$\boxed{V_2 = 320V}$

$V_3 \longrightarrow R_3$

$\Rightarrow V_3 = IR_3$

$= 0.016 \times 10,000$

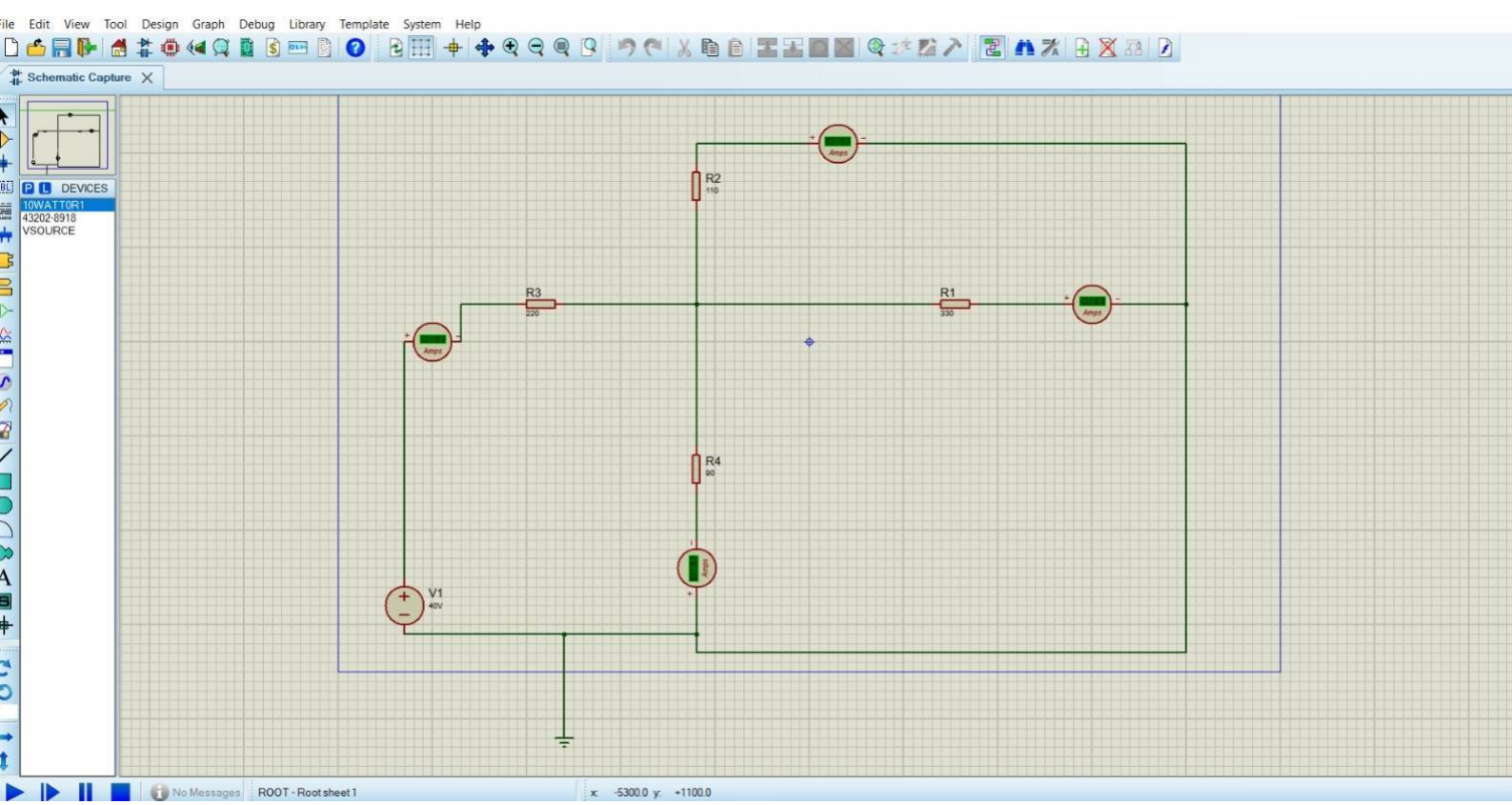
$\boxed{V_3 = 160V}$

Verify KVL

$V - V_1 - V_2 - V_3 = 0$

$800 - 320 - 320 - 160 = 0$

$\boxed{800 = 320V + 320V + 160V}$


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Schematic Capture
DEVICES
10WATT0R1
43202-8918
VSOURCE
R2
110
R3
220
R1
330
R4
90
V1
40V
Amps
No Messages
ROOT - Root sheet 1
x: -5300.0 y: +1100.0

KCL EXPERIMENT

MANUAL CHECKING

→ $V = 40V$
→ $R_1 = 330\Omega$
→ $R_2 = 110\Omega$
→ $R_3 = 220\Omega$
→ $R_4 = 90\Omega$

R_1, R_2, R_4 are in Parallel

using Kirchoff's Current law,

$$I_3 = I_2 + I_1 + I_4.$$

→ Ohm's law.
(Sub)

$$\frac{40 - V_A}{220} = \frac{V_A}{110} + \frac{V_A}{330} + \frac{V_A}{90}$$

$$\boxed{V_A = 6.545V}$$

$R_2 = 110\Omega$.

$$\Rightarrow I_2 = \frac{V_A}{R_2} = \frac{6.545}{110}$$

$$\boxed{I_2 = 0.0595 \approx 0.6A}$$

$R_1 = 330\Omega$

$$\Rightarrow I_1 = \frac{6.545}{330}$$

$$\boxed{I_1 = 0.0198A \approx 0.02A}$$

$R_4 = 90\Omega$

$$\Rightarrow I_4 = \frac{6.545}{90}$$

$$\boxed{I_4 = 0.0727A \approx 0.07A}$$

$R_3 = 220\Omega$

$$\Rightarrow I_3 = \frac{40 - 6.545}{220}$$

$$\boxed{I_3 = 0.1521A \approx 0.15A}$$

$$I_{in} = I_3 = 0.1521\,A$$

$$I_{out} = I_2 + I_1 + I_4$$

$$= 0.0595 + 0.0198 + 0.0727$$

$$I_{out} = 0.1520\,A$$

$$\therefore \boxed{I_{in} = I_{out}}$$